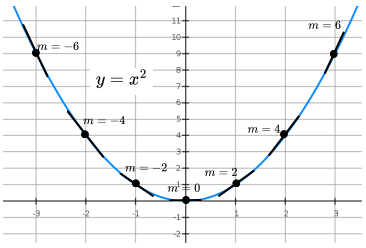
1.30 Differentiation from First Principles

Chapter 12 · Original slide 3 · Gradient Function



$m = 3$



At GCSE, you found the gradient of a curve at a particular point by drawing a tangent.

For a straight line, the gradient is constant:

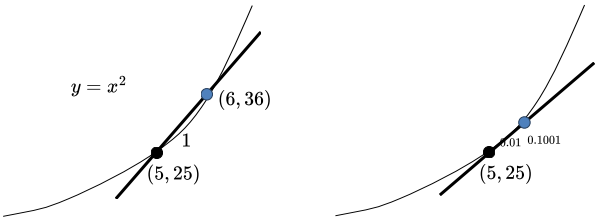
$y = 3x + 2$

However, for a curve the gradient varies. We can no longer have a single value for the gradient; we ideally want an expression in terms of $\mathbf{x}$ that gives us the gradient for any value of x (unsurprisingly known as the gradient function).

By looking at the relationship between x and the gradient at that point, can you come up with an expression, in terms of x for the gradient?

Gradient Function $=$

Chapter 12 · Original slide 4 · Finding the Gradient Function



The question is then: Is there a method to work out the gradient function without having to draw lots of tangents and hoping that we can spot the rule?

To approximate the gradient on the curve $y = x^2$ when $x = 5$, we could pick a point on the curve just slightly to the right, then find the gradient between the two points:

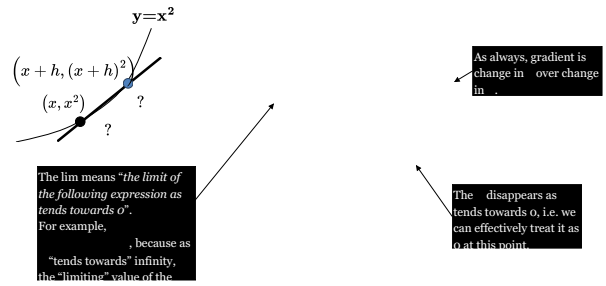
$(5.01, 25.1001)$

As the second point gets closer and closer, the gradient becomes a better approximation of the true gradient:

The actual gradient when $x = 5$ is 10, so this approximation is damn close!

1.30 Differentiation from First Principles

Chapter 12 · Original slide 5 · Finding the Gradient Function



This gives us a numerical method to get the gradient at a particular **x**, but doesn't give us the gradient function in general. Let's use exactly the same method, but keep x general, and make the 'small change' (which was previously 0.01) 'h':

And voila, we got the $2x$ we saw earlier!

Chapter 12 · Original slide 6 · Finding the Gradient Function

! The gradient function, or derivative, of the curve $y = f(x)$ is written as $f\prime(x)$ or $\frac{dy}{dx}$.

$f'(x) = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$

The gradient function can be used to find the gradient of the curve for any value of x.

We will soon see an easier/quicker way to differentiate expressions like $y = x^3 - x$ without using 'limits'. But this method, known as differentiating by first principles, is now in the A Level syllabus, and you could be tested on it!

Notation Note:

Whether we use $\frac{dy}{dx}$ or $f\prime(x)$ for the gradient function depends on whether we use $y =$ or $f(x) =$ to start with:

$y = x^2 \to \frac{dy}{dx} = 2x$
$f(x) = x^2 \to f'(x) = 2x$

"Leibniz's notation"

"Lagrange's notation"

Advanced Notation Note:

Rather than h for the small change in x, the formal notation is δx. So actually:

$f'(x) = \lim_{\delta x \to 0} \frac{f(x+\delta x)-f(x)}{\delta x}$

So we in fact have 3 symbols for "change in"!

- Δx: any change in x (as seen in Chp5: $m = \frac{\Delta y}{\Delta x}$)
- δx: a small change in x
- dx: an infinitesimally small change in x

1.30 Differentiation from First Principles

Chapter 12 · Original slide 7 · Example

The point A with coordinates $(4, 16)$ lies on the curve with equation $y = x^2$.

At point A the curve has gradient g.

a) Show that $g = \lim_{h \to 0} (8 + h)$

b) Deduce the value of g.

Use the “differentiation by first principles” formula.

b

Function is $f(x) = x^2$

Chapter 12 · Original slide 8 · Test Your Understanding

Prove from first principles that the derivative of x^4 is $4x^3$.

As $h \to 0$, any term involving a multiplication by h will become 0.

Support:

“Row 4” of Pascal’s Triangle is:

1 4 6 4 1

You’re welcome.

1.30 Differentiation from First Principles

Chapter 12 · Original slide 9 · Exercise 12B – Pearson Pure Y1 page 261-262

5 Prove, from first principles, that the derivative of $6x$ is 6.

Reminder:

$f\prime(x)$ and $\frac{dy}{dx}$ both mean the gradient function, also known as the derivative of y.

$f'(x) = \frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$

Chapter 12 · Original slide 10 · Exercise 12B – Pearson Pure Y1 page 261-262

6 Prove, from first principles, that the derivative of $4x^2$ is $8x$.

Reminder:

$f\prime(x)$ and $\frac{dy}{dx}$ both mean the gradient function, also known as the derivative of y.

$f'(x) = \frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$

1.30 Differentiation from First Principles

Chapter 12 · Original slide 11 · Exercise 12B – Pearson Pure Y1 page 261-262

7 $\mathrm{f}(x) = ax^2$, where a is a constant. Prove, from first principles, that $\mathrm{f}'(x) = 2ax$.

Reminder:

$f\prime(x)$ and $\frac{dy}{dx}$ both mean the gradient function, also known as the derivative of y.

$f'(x) = \frac{dy}{dx} = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$

Chapter 12 · Original slide 12 · Exercise 12B – Pearson Pure Y1 page 261-262

Challenge $\mathrm{f}(x) = \frac{1}{x}$.

a Given that $\mathrm{f}'(x) = \lim_{h\to 0} \frac{\mathrm{f}(x+h)-\mathrm{f}(x)}{h}$, show that $\mathrm{f}'(x) = \lim_{h\to 0} \frac{-1}{x^2+xh}$.

b Deduce that $\mathrm{f}'(x) = -\frac{1}{x^2}$.